

Field-Circuit Coupling Using Black-Box Simulation Tools

Coupling to a Reduced-Order Field Representation via Waveform Relaxation

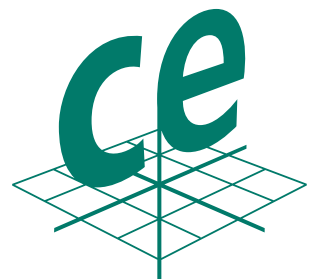
Bachelor thesis in the field of study "Computational Engineering" by Philip Simon

Date of submission: August 2, 2026

1. Review: Gutachter 1
2. Review: Gutachterin 2
Darmstadt



TECHNISCHE
UNIVERSITÄT
DARMSTADT



field of study:
Computational Engineering

Field-Circuit Coupling Using Black-Box Simulation Tools
Coupling to a Reduced-Order Field Representation via Waveform Relaxation

Bachelor thesis in the field of study “Computational Engineering” by Philip Simon

Date of submission: August 2, 2026

Darmstadt

Erklärung zur Abschlussarbeit gemäß § 22 Abs. 7 APB TU Darmstadt

Hiermit erkläre ich, Philip Simon, dass ich die vorliegende Arbeit gemäß § 22 Abs. 7 APB der TU Darmstadt selbstständig, ohne Hilfe Dritter und nur mit den angegebenen Quellen und Hilfsmitteln angefertigt habe. Ich habe mit Ausnahme der zitierten Literatur und anderer in der Arbeit genannter Quellen keine fremden Hilfsmittel benutzt. Die von mir bei der Anfertigung dieser wissenschaftlichen Arbeit wörtlich oder inhaltlich benutzte Literatur und alle anderen Quellen habe ich im Text deutlich gekennzeichnet und gesondert aufgeführt. Dies gilt auch für Quellen oder Hilfsmittel aus dem Internet.

Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Mir ist bekannt, dass im Falle eines Plagiats (§ 38 Abs. 2 APB) ein Täuschungsversuch vorliegt, der dazu führt, dass die Arbeit mit 5,0 bewertet und damit ein Prüfungsversuch verbraucht wird. Abschlussarbeiten dürfen nur einmal wiederholt werden.

Bei einer Thesis des Fachbereichs Architektur entspricht die eingereichte elektronische Fassung dem vorgestellten Modell und den vorgelegten Plänen.

Darmstadt, 2. August 2026

P. Simon

Abstract

Electric machines operate at relatively low frequencies yet are typically driven by high-frequency power-electronic circuitry, so they are most efficiently time-integrated at different rates. Their differing character – one a complex machine, modelled as a distributed device with differential equations and simulated numerically using methods such as the Finite Element Method (FEM), the other a circuit solved by its own solver – motivates a partitioned co-simulation rather than a monolithic solve. This work couples the domains with a Waveform Relaxation (WR) scheme, in which an inductance-based reduced-order model of the field serves as the interface condition and, as a linearisation term, accelerates convergence. The scheme is implemented with Xyce, Sandia's open-source SPICE simulator, leveraging its checkpoint/restart feature to re-solve each time window across WR iterations. The implementation is verified against monolithic solutions for cases with varying circuit complexity and a dummy FEM solver. A parameter study over [window length/linearisation update / ...] shows that [key result], from which [a concrete guideline, e.g. for choosing the window size] is derived.

Contents

1	Einleitung	1
1.1	Motivation	1
1.2	State of the Art	1
1.3	Objective and Contributions	2
1.4	Limitations	3
1.5	Outline of the Thesis	3
2	Fundamentals	4
2.1	Modified Nodal Analysis	4
2.2	Dynamic Iteration	4
2.3	Coupling Scheme	5
3	Coupling with Xyce™	7
3.1	Requirements on the Circuit Solver (Why Xyce?)	7
3.2	Field Dummy Solver	7
3.3	Coupling via the Xyce Binary	7
3.3.1	Restarts and Checkpointing	7
3.3.2	Interface Condition via Behavioural Source	7
3.3.3	Coupling Direction	7
3.4	General External Interface and Its Limitations	7
4	Benchmark and Verification	8
5	Application to General Circuits	9
6	Conclusions and Outlook	10

1 Einleitung

1.1 Motivation

Most parts of a drive system, such as sources, switches, passive elements, and control circuitry, do not need to be modelled with the full spatial detail of Maxwell's partial differential equations [11] and can be adequately described by lumped elements. These reduced models are formulated as differential-algebraic equations (DAEs), e.g. by modified nodal analysis (MNA) [5]. For the machine itself, a lumped representation may be insufficient whenever saturation, eddy-current effects or the spatial distribution of the field govern its behaviour. A spatially distributed model is used instead, commonly obtained from the magnetoquasistatic (MQS) approximation of Maxwell's equations and discretised in space, e.g. by the finite element method (FEM) [6]. Coupling such a field model to the circuit yields a system of partial differential-algebraic equations (PDAEs), which reduces to a system of DAEs once the field model is discretised in space [2]. Solving the system *monolithically* would require a single time integrator to resolve the dynamics of the most active domain, so the latent domain is advanced with a step size far smaller than its dynamics require. Partitioning removes this restriction, as each subsystem can be integrated with a step size adapted to its own time constants. This is particularly beneficial when the time scales of the domains are separated by orders of magnitude, where it has been shown to reduce the computational effort considerably [10]. Electric machines are increasingly fed by fast-switching power electronics, enabled by wide-bandgap semiconductor devices and driven by ambitious power-density and voltage targets for the coming decade, see Tab. 1.1. In such systems, the machine is the latent subsystem, as its electromagnetic and mechanical time constants exceed the converter switching period by orders of magnitude, which motivates multirate and co-simulation approaches [9, 7]. Faster switching also increases the voltage slew rate dv/dt at the machine terminals. Whereas insulation systems of standard machines are designed for slew rates of a few $\text{kV}\mu\text{s}^{-1}$, silicon-carbide devices reach values exceeding $100\text{ kV}\mu\text{s}^{-1}$ [8]. Consequently, the behaviour of the machine at its terminals is no longer adequately represented by a lumped element, motivating a partitioned field/circuit approach in which the interface conditions capture the machine's terminal response.

Practically speaking, there are well-established solvers for each domain. However, commercial software is often closed-source and access to the internal variables during simulation is limited, making coupling the field/circuit domains difficult, cf. [4]. Hence, a so-called *black-box* view of the problem, which is agnostic to the solvers, is the versatile implementation. More precisely, one might call it a *grey-box* view, since we require some knowledge of the state of the field domain in the interface condition, as we will discuss later.

1.2 State of the Art

Solving the system of DAEs is a thoroughly studied problem in the context of simulating field/circuit domains. Approaches differ in how tightly the subsystems are coupled, ranging from monolithic solves, where all

Table 1.1: Power electronics and electric motor targets (US DoE).

	2025	2030	2035
Voltage (V)	600	800	800
Power Density Electronics (kW/L)	150	200	225
Power Density Machines (kW/L)	75	100	112,5

Daten aus [12].

equations are assembled into a single system, to loosely coupled *dynamic iteration* (also called *Waveform Relaxation* (WR)) where the subsystems are solved separately and exchange waveforms over time windows, see Section ???. The monolithic approach requires a shared solver for both domains, which conflicts with the black-box setting and is prohibitively expensive in computation for the targeted application here, see Section 1.1. In the case of dynamic iteration, it has been identified that the exchange of algebraic variables is a mathematical cause for divergence [9]. Thus, enhanced modelling of the interface via a *surrogate* or *reduced order model* (ROM) has been shown to improve convergence significantly. Without it, the iteration may be divergent [1]. While such an interface condition is straightforward to formulate analytically, its time discretisation involves the time integrators of both subsystems. When black-box tools are coupled, each subsystem discretises the shared condition with its own integrator, so the two discrete operators may differ. Different operators result in an inconsistent interface condition and may lead to a false fixed point or a divergent iteration, as we will see in Chapter 4. This motivates the need for a consistent implementation of the interface without access to the solver’s time integrator. Xyce™ also provides a native coupling interface (General External Interface). Why it is not used for the dynamic iteration here is discussed in Section 3.4.

1.3 Objective and Contributions

The contributions of this thesis are:

1. a dynamic iteration co-simulation of the field/circuit built entirely from black-box, netlist-driven simulators (demonstrated with Xyce™);
2. a consistent formulation and implementation of the interface condition that does not require access to either solver’s time integrator;
3. a study of the resulting scheme’s convergence behaviour, transmission conditions and parameter-dependence, e.g. window-length, isolated from any particular field discretisation.

Xyce™ is the open-source analogue circuit simulator used in this thesis. It can be replaced by other netlist-driven simulators with little change to the implementation, provided they offer:

- Restart/Checkpointing capability
- Behavioural source-elements

1.4 Limitations

The coupling scheme developed in this work makes no assumption about how the field model is formulated or discretised internally, at the cost of not demonstrating the method on an industrial machine model. The latter is left to future work. Considering anything other than two-port distributed field devices would require substantial changes to the codebase. The ROM values are constant across the simulation, such that changes of state in the distributed device, e.g. by saturation, are not accurately modelled by the ROM. The implementation only covers transient simulation.

1.5 Outline of the Thesis

In Chapter 2, the fundamental theory that is required to explain the implementation and discuss any results is laid out. Chapter 3 documents the implementation of the co-simulation with the Xyce[™] parallel electronic simulator. In Chapter 4, the implementation is verified against a suite of benchmark problems. Chapter 5 contains parameter studies and a general approach to choosing them. Chapter 6 concludes the thesis and outlines possibilities for future work.

2 Fundamentals

One of the most significant ideas that improved simulation speed by up to two orders of magnitude is decomposing large systems into coupled subsystems that can be integrated independently [4]. To implement the coupling of the field and circuit domains via dynamic iteration, the underlying concepts need to be explained first.

2.1 Modified Nodal Analysis

The equations that govern the behaviour of the circuit subsystem can be written in abstract form as

$$\mathbf{M} \frac{d}{dt} \mathbf{x}(t) + \mathbf{K} \mathbf{x}(t) = \mathbf{r}(t) \quad (2.1a)$$

$$\mathbf{v}_C(t) = \mathbf{P}^\top \mathbf{x}(t) \quad (2.1b)$$

where, in case of MNA, $\mathbf{x}$ denotes the unknown functions of node potentials and the voltage-source and inductor branch currents

$$\mathbf{x}^\top(t) = [\varphi^\top(t) \quad \mathbf{i}_L^\top(t) \quad \mathbf{i}_V^\top(t)].$$

And $\mathbf{v}_C$ is the voltage at the field-circuit interface and $\mathbf{P}$ is the according port-operator.

2.2 Dynamic Iteration

The subsystems are integrated independently for a period of time $T_n \rightarrow T_{n+1}$, where $T_n = [t_n, t_{n+1}]$, using the outputs of neighbouring subsystems as inputs. This is the essence of dynamic iteration [4]. It is a fixed-point iteration technique, i.e., the iteration

$$x^{k+1} = \varphi(x^k), \quad k = 0, 1, 2, \dots$$

converges to a fixed point, where $x^* = \varphi(x^*)$.

In the case of a field-circuit coupling, the vector of unknowns x contains the voltage (node potentials) and (branch) current across the interface ports. The fixed point is reached if the voltages and currents of both subsystems converge, i.e.,

$$\begin{array}{ll} \mathbf{v}_C^{k+1} = \mathbf{v}_C^k & \mathbf{v}_F^{k+1} = \mathbf{v}_F^k \\ \underbrace{\mathbf{i}_C^{k+1} = \mathbf{i}_C^k}_{\text{circuit}} & \underbrace{\mathbf{i}_F^{k+1} = \mathbf{i}_F^k}_{\text{field}} \end{array} \quad (2.2)$$

where V_C and I_C are the port voltage and current of the circuit-side, and V_F and I_F are the port voltage and current of the field-side.

2.3 Coupling Scheme

Transmission between the subsystems is handled at the ports, i.e., the inputs and outputs of a subsystem. Two independent factors determine the coupling of the field and circuit: the coupling direction and the transmission condition itself.

Coupling Direction The direction of coupling stipulates which variable from one subsystem (output) is prescribed on another (input) during iteration. For our field/circuit co-simulation, it answers the question: *“Does the field ingest a current or a voltage as its (plain) input?”*. This consideration directly impacts the index of the resulting DAE. It has been shown that voltage-driven field formulations (A^* and $T\text{-}\Omega$) result in a lower index system (index-1) and thereby offer better numerical stability than the current-driven alternative (index-2) [2]. Hence, a voltage-driven coupling

$$v_F \leftarrow v_C$$

is likely the preferred choice for this work’s implementation, see Section 3, though under some scopes the current-driven approach may be considered, cf. e.g. [3].

Transmission Conditions The transmission conditions, also called interface conditions, specify *how* the exchanged variables enter each subsystem as inputs and leave it as outputs. For a voltage-driven coupling, the plain (zero-order) conditions of the field/circuit problem read

$$v_F \leftarrow v_C \quad \text{and} \quad i_C \leftarrow i_F, \quad (2.3)$$

that is, the field is prescribed the voltage and returns the current. The exchanged quantities are waveforms, $v_C = v_C(t)$ and $i_F = i_F(t)$, which gives the method its name.

Extracting a reduced-order model (ROM) of the field domain improves the coupling, and the better the ROM, the larger the admissible time windows [10, 9]. Because the spatially discretised magnetoquasistatic field behaves as an inductance-like element [2], the resulting ROM is itself inductive. Later work established that this inductance is precisely the first-order term of a Neumann-series expansion of the optimal transmission condition [3].

Following the suggestion of [3] to excite the field by voltages rather than currents, the optimal transmission condition follows by an analogous derivation.

Let the field problem be voltage driven within a GAUSS–SEIDEL scheme. The zero-order Dirichlet conditions read

$$v_F^{k+1} = v_C^{k+1}, \quad (2.4a)$$

$$i_C^{k+1} = i_F^k. \quad (2.4b)$$

The circuit-side condition (2.4b) is generalised to a linear combination of current and voltage,

$$i_C^{k+1} - \beta v_C^{k+1} = i_F^k - \beta v_F^k, \quad (2.5)$$

where β is the weighting factor. It is worth noting, that the field's input condition (2.4a) is left plain: unlike the field, an arbitrary circuit does not admit an easily extractable impedance, so its optimisation is cumbersome, cf. [3].

Eliminating the internal degrees of freedom from the frequency-domain circuit equations yields the port relation $V_C = g_c - \mathbf{x}_P I_C$, with transfer impedance $\mathbf{x}_P(\omega) = \mathbf{P}^\top (\mathbf{A}_c j\omega + \mathbf{B}_c)^{-1} \mathbf{P}$ and projected source term $g_c = P^\top (\mathbf{A}_c j\omega + \mathbf{B}_c)^{-1} g(\omega)$. The voltage-driven field returns $I_F = \mathbf{Y} V_F$, where the admittance $\mathbf{Y}(\omega) = \mathbf{Z}^{-1}(\omega)$ is the inverse of the port impedance $\mathbf{Z}(\omega)$ of the MQS problem [3]. Inserting the field response and the plain condition (2.4a) into (2.5) and eliminating V_C^{k+1} gives the error propagation

$$i_C^{k+1} - i_C^k = \rho(\beta) (i_C^k - i_C^{k-1}), \quad \rho(\beta) = (\mathbf{I} + \beta \mathbf{x}_P)^{-1} (\beta - \mathbf{Y}) \mathbf{x}_P.$$

In order for the fixed point iteration to converge, the error across iterations has to contract, i.e., $|\rho(\beta)| < 1$. Since $\mathbf{x}_P$ is invertible, optimal convergence $\rho(\beta) = 0$ is attained for

$$\beta = \mathbf{Y}(\omega),$$

so that the optimised circuit-side transmission condition reads

$$i_C^{k+1} - \mathbf{Y}(\omega) v_C^{k+1} = i_F^k - \mathbf{Y}(\omega) v_F^k. \quad (2.6)$$

Since $\mathbf{Z} \approx j\omega \mathbf{L}$ is the low-frequency approximation of the exact port impedance (leading term of the Neumann series, corresponding to the magnetostatic inductance), the admittance inherits $\mathbf{Y} \approx (j\omega \mathbf{L})^{-1}$ in the same regime. As in the current-driven case, this frequency-domain expression must be approximated in our time-domain implementation, where $j\omega$ corresponds to a differentiation in time [3]. The resulting condition,

$$i_F(t) \approx \mathbf{L}^{-1} \int v_F(t) dt$$

represents the dual of the flux-linkage condition of the current-driven scheme. Since the underlying inductance is relatively cheap to extract (a single magnetostatic solve), it could be re-evaluated at successive working points throughout the transient to track the magnetic saturation effect of the field subsystem, at the cost of repeated extraction; this is not implemented in this work. Eddy-current effects are not covered in this first-order approximation of the inductance – those dynamics are compensated by the lagged correction term $\Delta i_F^k := i_F^k - \mathbf{Y}(\omega) v_F^k$ in Equation (2.6)

Note on Algorithms Though we used a GAUSS–SEIDEL scheme (Einzelschrittverfahren) to obtain a solution of the DAEs in this implementation, other algorithms such as GAUSS–JACOBI (Gesamtschrittverfahren) are common, too. While the former offers faster convergence, the latter has better parallelizability [1].

3 Coupling with Xyce™

To successfully implement dynamic iteration, which is a fixed-point iteration technique, a rollback to repeat the co-simulation step with corrected inputs is required.

3.1 Requirements on the Circuit Solver (Why Xyce?)

3.2 Field Dummy Solver

3.3 Coupling via the Xyce Binary

3.3.1 Restarts and Checkpointing

3.3.2 Interface Condition via Behavioural Source

3.3.3 Coupling Direction

3.4 General External Interface and Its Limitations



4 Benchmark and Verification

Test Suite.



5 Application to General Circuits

Parameterstudien und -Empfehlungen.



6 Conclusions and Outlook

Zusammenfassung der Ergebnisse, Schlussfolgerung und Ausblick. Keine neuen Informationen mehr.



List of Figures



List of Tables

1.1	Power electronics and electric motor targets (US DoE).	2
-----	--	---

Bibliography

- [1] Andreas Bartel et al. “Dynamic Iteration for Coupled Problems of Electric Circuits and Distributed Devices”. In: *SIAM Journal on Scientific Computing* 35.2 (2013), B315–B335. DOI: 10.1137/120867111. eprint: <https://doi.org/10.1137/120867111>. URL: <https://doi.org/10.1137/120867111>.
- [2] Idoia Cortes Garcia, Herbert De Gersem, and Sebastian Schöps. “A structural analysis of field/circuit coupled problems based on a generalised circuit element”. In: *Numerical Algorithms* 83.1 (Jan. 1, 2020), pp. 373–394. ISSN: 1572-9265. DOI: 10.1007/s11075-019-00686-x. URL: <https://doi.org/10.1007/s11075-019-00686-x>.
- [3] Idoia Cortes Garcia et al. “Optimized Field/Circuit Coupling for the Simulation of Quenches in Superconducting Magnets”. In: *IEEE Journal on Multiscale and Multiphysics Computational Techniques* 2 (2017), pp. 97–104. DOI: 10.1109/JMMCT.2017.2710128.
- [4] Cláudio Gomes et al. *Co-simulation: State of the art*. 2017. URL: <https://arxiv.org/abs/1702.00686>.
- [5] Chung-Wen Ho, A. Ruehli, and P. Brennan. “The modified nodal approach to network analysis”. In: *IEEE Transactions on Circuits and Systems* 22.6 (1975), pp. 504–509. DOI: 10.1109/TCS.1975.1084079.
- [6] Peter Monk. *Finite Element Methods for Maxwell’s Equations*. Oxford University Press, Apr. 2003. ISBN: 9780198508885. DOI: 10.1093/acprof:oso/9780198508885.001.0001. URL: <https://doi.org/10.1093/acprof:oso/9780198508885.001.0001>.
- [7] Andreas Pels et al. “Efficient Simulation of DC–DC Switch-Mode Power Converters by Multirate Partial Differential Equations”. In: *IEEE Journal on Multiscale and Multiphysics Computational Techniques* 4 (2019), pp. 64–75. DOI: 10.1109/JMMCT.2018.2888900.
- [8] Timo Petri, Marina Keller, and Nejila Parspour. “The Insulation Resilience of Inverter-Fed Low Voltage Traction Machines: Review, Challenges, and Opportunities”. In: *IEEE Access* 10 (Jan. 2022), pp. 104023–104049. DOI: 10.1109/ACCESS.2022.3210348.
- [9] Sebastian Schöps. “Multiscale modeling and multirate time integration of field/circuit coupled problems”. PhD thesis. Bergische Universität Wuppertal, 2011. URL: <https://nbn-resolving.org/urn:nbn:de:hbz:468-20110718-110524-8>.
- [10] Sebastian Schops, Herbert De Gersem, and Andreas Bartel. “A Cosimulation Framework for Multirate Time Integration of Field/Circuit Coupled Problems”. In: *IEEE Transactions on Magnetics* 46.8 (2010), pp. 3233–3236. DOI: 10.1109/TMAG.2010.2045156.
- [11] Karl Simonyi. *Theoretische Elektrotechnik*. 6th ed. Berlin: VEB Deutscher Verlag der Wissenschaften, 1977.
- [12] U.S. DRIVE Electric Drive Technical Team. *Electric Drive Technical Team Roadmap*. Tech. rep. U.S. Department of Energy, Vehicle Technologies Office, Mar. 2024. URL: https://www.energy.gov/sites/default/files/2024-06/EDTT_Roadmap_2023_JOG_Consensus_compliant.pdf (visited on 07/23/2026).